The Discriminant Resolvent of the A_5-Cubic Pencil

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Abstract

The nonstandard A_5-invariant pencil of cubic threefolds carries five principal gluing kernels: three rational and two golden. We prove that this is not an accidental $3+2$ count. The five kernels are the two proper transitive resolvents of the S_3-torsor of ordered nonzero two-torsion on the relative elliptic norm axis. The rational triple is the root cover and the golden pair has the sign character, hence is the discriminant double cover of the same two-division cubic.

Writing t for the signed cubic parameter and $T = 81t^2$, the unmarked elliptic period is the standard $X_0(3)$-coordinate

$$j = \frac{(T+27)(T+3)^3}{T}.$$

The signed cubic line is the sign modular curve $r^2 = T$, with $r = 9t$ after choosing its deck convention. The degree-three resolvent is $X_0(6)$, and their common degree-six Galois closure belongs to $\Gamma_0(3) \cap \Gamma(2)$. We give rational equations for all three covers and distinguish the two modular cusps from the additional interior boundary points where the cubic degenerates. In particular the modular identification concerns the actual relative principal-kernel packet, not only an auxiliary elliptic curve with the same j-invariant.

Keywords. cubic threefold; intermediate Jacobian; elliptic two-torsion; discriminant resolvent; modular curve; A_5.

1 The result

Let $W \subset \mathbf{C}^6$ be the hyperplane $\sum_{i=0}^{5} z_i = 0$. The nonstandard transitive copy of $A_5 \subset S_6$ has two orbits $\mathcal{T}_+$ and $\mathcal{T}_-$, each of size ten, on the three-element subsets of $\{0, \dots, 5\}$. Put

$$P = \sum_{i=0}^{5} z_i^3, \qquad Q = \sum_{I \in \mathcal{T}_+} \prod_{i \in I} z_i - \sum_{I \in \mathcal{T}_-} \prod_{i \in I} z_i, \qquad X_t = \{P + tQ = 0\} \subset \mathbf{P}(W).$$

The outer coset of the normalizer $S_5 = N_{S_6}(A_5)$ fixes P, negates Q, and therefore sends X_t to X_{-t}. Thus t is the signed parameter and t^2 is the unmarked parameter.

On the smooth marked base, the six distinguished dihedral axes in the intermediate Jacobian form one relative elliptic norm axis $\mathcal{E}$. The integral six-axis construction of [4, Relative six-axis source and Principal gluing packet] identifies the five principal kernels with the finite local system

$$\mathscr{P} = PP_{\mathbf{F}_4}(\mathbf{F}_4 \otimes_{\mathbf{F}_2} \mathcal{E}[2]). \tag{1.1}$$

The comparison in Proposition 3.1 below records why the elliptic curve used to compute the period is this actual norm axis at the level needed in (1.1).

Theorem 1.1 (Cubic–modular resolvent)**.** *Set $T = 81t^2$. Then the following assertions hold over the smooth cubic base.*

1. *The elliptic norm axis has*

$$j(\mathcal{E}) = J(T) := \frac{(T+27)(T+3)^3}{T}.$$

   Consequently the unmarked period map identifies its normalized coarse base with an open subcurve of $X_0(3)$.

2. *The packet $\mathscr{P}$ is the disjoint union of the root cover $\mathbf{P}(\mathcal{E}[2])$, of degree three, and a degree-two cover having the sign character of $\mathrm{Gal}(\mathcal{E}[2]) \subset S_3$. The second cover is the discriminant resolvent of the two-division cubic.*

3. *Over $X_0(3)$, the root cover is $X_0(6)$, the sign cover has function field $\mathbf{Q}(T)(\sqrt{T})$, and their common Galois closure belongs to $\Gamma_0(3) \cap \Gamma(2)$. After one choice of deck involution, the signed cubic coordinate identifies with the sign coordinate by*

$$r^2 = T, \qquad r = 9t.$$

4. *The compact sign curve has genus zero, two cusps of widths 2 and 6, and two elliptic points of order three. Its map to $X_0(3)$ is ramified at the two cusps and nowhere else as a coarse map.*

5. *The geometric mod-two monodromy of the actual norm axis over the marked smooth cubic base is exactly $A_3 \simeq C_3$.*

The smooth cubic base is obtained from the sign modular curve by deleting, in addition to the cusps over $T = 0, \infty$, the points over $T = -27$ and $T = 729/5$. Those last two values are interior modular points, not cusps.

The theorem identifies local systems and their monodromy. It does not say that a fivefold gluing is literally an elliptic 2-isogeny, nor that the compactified cubic pencil is the modular stack.

2 The finite $3 + 2$ resolvent

Let V be a two-dimensional $\mathbf{F}_2$-vector space and $V_4 = \mathbf{F}_4 \otimes_{\mathbf{F}_2} V$. Scalar extension gives a distinguished three-element subset

$$\mathbf{P}_{\mathbf{F}_2}(V) \subset \mathbf{P}_{\mathbf{F}_4}(V_4).$$

Write $D(V)$ for its two-element complement.

Lemma 2.1 (Sign lemma)**.** *Under $\mathrm{GL}(V) \simeq S_3$, the action on $D(V)$ is the sign of the natural action on $\mathbf{P}(V)$. Thus the two finite S_3-sets are*

$$\mathbf{P}(V) \simeq S_3/C_2, \qquad D(V) \simeq S_3/A_3.$$

Proof. Choose a basis and write

$$\mathbf{P}^1(\mathbf{F}_4) = \{\infty, 0, 1, \omega, \omega^2\}, \qquad \mathbf{P}^1(\mathbf{F}_2) = \{\infty, 0, 1\}.$$

The transformations $x \mapsto x + 1$ and $x \mapsto 1/(x + 1)$ generate $\mathrm{PGL}_2(\mathbf{F}_2)$. The first acts as $(0\ 1)(\omega\ \omega^2)$; the second cycles $(\infty\ 0\ 1)$ and fixes ω, ω^2. The first generator is odd on each of the three-set and the two-set, while the second is even on both. This proves the assertion. $\square$

For a degree-three finite *étale* cover $C \to S$, let $\mathrm{Or}(C/S)$ denote the double cover associated to the sign of its monodromy in S_3. Lemma 2.1 proves

$$[D(V)] = [\mathrm{Or}(\mathbf{P}(V)/S)] \quad \text{in } H^1(S, C_2). \tag{2.1}$$

This equality of torsor classes is canonical. A sheetwise isomorphism of the two covers has two choices, interchanged by the deck involution. This minor distinction is what makes the normalization $r = 9t$, rather than $r = -9t$, a convention.

Now let an elliptic curve be given locally, with 2 invertible, by

$$E : y^2 = x^3 + Ax + B.$$

The nonzero two-torsion points are the roots of $f_2(x) = x^3 + Ax + B$. The orientation cover of this root cover is therefore

$$u^2 = \mathrm{disc}(f_2), \qquad \mathrm{disc}(f_2) = -4A^3 - 27B^2.$$

Since $\Delta_E = 16\,\mathrm{disc}(f_2)$, it has the same square class as the elliptic discriminant. Invariantly, Δ_E is a nowhere-vanishing section of the twelfth power of the Hodge line; the double cover is the torsor of its square roots. No global scalar equation for Δ_E is implied before a Hodge trivialization is chosen.

Applying the lemma to (1.1) proves Theorem 1.1(2). It also gives a useful monodromy consequence: choosing one golden sheet reduces the mod-two image from S_3 to $A_3 \simeq C_3$.

3 The cubic parameter and the actual norm axis

The period calculation starts with the Prym quotient used by van Geemen and Yamauchi. In their coordinate u, two auxiliary parameters in the Weierstrass model of its elliptic factor are

$$a(u) = -\frac{32(u-3)^4}{9(u+1)^3(u+3)^2}, \qquad b(u) = -\frac{8(u-3)^2(u-1)}{(u+1)(u+3)^2}.$$

In the notation of [5, Proposition 3.2], the generalized Weierstrass coefficients are

$$\begin{aligned} a_1 &= b^2/4 + b - 4, & a_3 &= (a+b)^2/2 + 4a + 6b - 8,\\ a_2 &= 2 - a - ab/2 - b^4/64 - b^3/8 - b^2/4 - b, & a_4 &= 4(a+b-1),\\ a_6 &= (a+b-1)(8 - 4a - 2ab - b^4/16 - b^3/2 - b^2 - 4b). \end{aligned}$$

The Fourier change of coordinates from the six-point equation gives $u = ct$, where $c^2 = 5$. Substitution in the standard formulas for c_4 and Δ, followed by $u^2 = 5t^2$, yields

$$j = \frac{9(3t^2+1)(27t^2+1)^3}{t^2}. \tag{3.1}$$

This calculation is replayed exactly in the paper's verification bundle.

Putting $T = 81t^2$, equation (3.1) becomes

$$j = J(T) = \frac{(T+27)(T+3)^3}{T}. \tag{3.2}$$

This is the standard degree-four j-map on $X_0(3)$. A convenient Tate normal form is

$$E_T : \quad y^2 + (T+27)xy + (T+27)^2 y = x^3. \tag{3.3}$$

The point $(0,0)$ has order three, and the standard Weierstrass formulas give

$$\Delta(E_T) = T(T+27)^8, \qquad j(E_T) = J(T). \tag{3.4}$$

The equality of j-maps alone would identify only a generic elliptic isomorphism class. The next proposition supplies the geometric comparison needed for the principal-kernel packet.

Proposition 3.1 (Actual-axis comparison)**.** *Over the common marked smooth A_5/D_5 base, the van Geemen–Yamauchi Prym factor maps isomorphically, as a polarized elliptic scheme, to the primitive dihedral norm axis inside the intermediate Jacobian. In particular its two-torsion local system is the $\mathcal{E}[2]$ occurring in* (1.1)*. Relative to* (3.3)*, it may differ by the quadratic twist*

$$D(T) = (T+27)(T-729/5),$$

which does not alter its two-torsion.

Proof. The degree-five cyclic quotient construction of [5, Propositions 2.1, 3.1, and 3.2] gives the Prym elliptic curve and a pullback map to the corresponding norm-fixed elliptic subvariety of the intermediate Jacobian. Pullback followed by norm is multiplication by five, so the image is the primitive one-dimensional fixed factor. On the Prym, the principal polarization pulls back as five times its intrinsic principal polarization. The restriction of the intermediate-Jacobian polarization to the primitive norm axis is likewise five times its primitive principal polarization. Factoring pullback through that axis and comparing degrees therefore gives degree one. This proves the polarized isomorphism.

The explicit change to (3.3) gives the displayed twist; alternatively, comparing their c_4, c_6 invariants gives the same square class. In characteristic different from two, a quadratic twist changes the Galois representation by a scalar $\{\pm 1\}$; that scalar is trivial on two-torsion and preserves every cyclic order-three subgroup. Hence either comparison identifies the required two-division local system, and the cyclic level-three marking descends even when a chosen generator does not. □

The proof also explains the role of earlier results. Roulleau exhibits the nonstandard A_5-pencil and its elliptic-power decomposition [3, Theorem 11(D) and Lemma 17]; Hartlieb places its intermediate Jacobians in a one-dimensional special subvariety [1, Lemma 5.5 and Proposition 5.7]. Neither statement by itself identifies the relative two-division packet used here.

4 The modular S_3-diagram

Reduction modulo two gives

$$\rho_2 : \Gamma_0(3) \longrightarrow \mathrm{SL}_2(\mathbf{F}_2) \simeq S_3.$$

It is surjective: the elements

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$$

belong to $\Gamma_0(3)$ and reduce to two distinct transvections, which generate $\mathrm{SL}_2(\mathbf{F}_2)$. Define

$$\Gamma_{\mathrm{sgn}}(6) = \ker(\mathrm{sgn} \circ \rho_2), \qquad \Gamma(2,3) = \Gamma(2) \cap \Gamma_0(3).$$

The first has index two and the second index six in $\Gamma_0(3)$. The resulting normalized Cartesian square is

$$\begin{array}{ccc} X(\Gamma(2,3)) & \xrightarrow{2} & X_0(6) \\ {\scriptstyle 3}\Big\downarrow & & \Big\downarrow{\scriptstyle 3} \\ X_{\mathrm{sgn}}(6) & \xrightarrow{2} & X_0(3). \end{array} \tag{4.1}$$

A nonzero point of $E[2]$ is the same as a cyclic subgroup of order two. Together with the given cyclic subgroup of order three, it determines a unique cyclic subgroup of order six. Conversely, an order-six cyclic subgroup has unique subgroups of orders two and three. Therefore the degree-three root resolvent is

$$X_0(6) \longrightarrow X_0(3).$$

The sign resolvent belongs to $\Gamma_{\mathrm{sgn}}(6)$, and the full ordered two-torsion cover belongs to $\Gamma(2,3)$. This proves the group part of Theorem 1.1(3).

All three covers are rational. With y on the root cover, r on the sign cover, and v on the common splitting cover, one choice of equations is

$$T = -\frac{(4y+3)(y+3)^2}{(y+1)^2}, \tag{4.2}$$

$$x = -\frac{4y^4}{(y+1)^2}, \tag{4.3}$$

$$r^2 = T, \tag{4.4}$$

$$y = -\frac{v^2+3}{4}, \qquad T = \frac{v^2(9-v^2)^2}{(1-v^2)^2}, \qquad r = \frac{v(9-v^2)}{1-v^2}. \tag{4.5}$$

Substitution of (4.2) and (4.3) in the two-division cubic below gives zero, so y really selects a nonzero two-torsion point. The last line directly verifies the root and sign cover equations. It realizes the classical fact that adjoining a root of an S_3-cubic and adjoining the square root of its discriminant generate its splitting field.

Corollary 4.1 (Golden orientation cyclicizes the rational packet)**.** *After base change to the sign curve, the root cover $X_0(6) \to X_0(3)$ becomes a connected Galois cover of degree three with group $A_3 \simeq C_3$. It is the full splitting cover of the two-division cubic. Hence, over the signed smooth cubic base, the rational three-sheet gluing packet has cyclic monodromy and any one rational sheet determines the other two.*

Proof. The sign subgroup has mod-two image A_3, while a point stabilizer in S_3 has order two. Their intersection is trivial. Therefore the pullback of the point-stabilizer cover to the sign cover has regular A_3-action on its three sheets. Equivalently, the fibre product of (4.2) and (4.4) is the splitting field parametrized by (4.5). □

For completeness, the two-division cubic of (3.3), obtained by eliminating y from the equation and $2y + (T+27)x + (T+27)^2 = 0$, is

$$4x^3 + (T+27)^2x^2 + 2(T+27)^3x + (T+27)^4.$$

Its discriminant is

$$16T(T+27)^8. \tag{4.6}$$

Thus its discriminant square class is exactly T. By Proposition 3.1 and Lemma 2.1, the exotic pair in the actual gluing packet is (4.4). Pulling back along $T = 81t^2$ splits it; choosing one sheet gives $r = 9t$. This completes the proof of Theorem 1.1(1)–(3). Over the sign curve the mod-two image is A_3, by construction of $\Gamma_{\mathrm{sgn}}(6)$. Passing to the smaller smooth cubic open cannot decrease that image: concretely, a small loop around either deleted lift of the order-three point $T = -27$ acts as a three-cycle. Proposition 3.1 transports this action to the actual norm axis. This proves assertion (5).

5 Compactification and the cubic boundary

The coarse orbifold $X_0(3)$ has two cusps of widths one and three and one elliptic point of order three. A primitive parabolic at either cusp reduces modulo two to a transposition, so it does not lie in $\Gamma_{\mathrm{sgn}}(6)$, while its square does. Each cusp consequently has one ramified lift, of width two or six. An order-three elliptic generator reduces to a three-cycle and lies in the sign kernel; the elliptic point has two unramified lifts, both of order three.

The sign subgroup has index eight in $\mathrm{PSL}_2(\mathbf{Z})$. The genus formula now reads

$$g = 1 + \frac{8}{12} - \frac{0}{4} - \frac{2}{3} - \frac{2}{2} = 0.$$

This proves Theorem 1.1(4), including the absence of any further coarse branching.

The same inertia calculation completes the passport of (4.1). On the root cover, a transposition has orbits of sizes one and two, giving cusp widths $1, 2, 3, 6$. On the regular split cover it has three orbits of size two, giving widths $2, 2, 2, 6, 6, 6$. The root and split subgroups contain no elliptic torsion. Their indices in $\mathrm{PSL}_2(\mathbf{Z})$ are twelve and twenty-four, so the genus formula gives genus zero in both cases. Thus every curve in the resolvent square is rational, now independently of the displayed parametrizations.

The eta Hauptmodul

$$h(\tau) = \left(\frac{\eta(\tau)}{\eta(3\tau)}\right)^{12}$$

has a simple pole at the width-one cusp and a simple zero at the width-three cusp. In the normalization (3.2),

$$h = \frac{729}{T}.$$

Because $729 = 27^2$, the quadratic fields $\mathbf{Q}(h)(\sqrt{h})$ and $\mathbf{Q}(T)(\sqrt{T})$ coincide. Analytically one may take $\sqrt{h} = (\eta(\tau)/\eta(3\tau))^6$. Its half-integral order in the base cusp parameter is the width doubling just computed.

Combining this square root with $r = 9t$ gives an explicit modular uniformization of the signed cubic parameter:

$$\boxed{t(\tau) = 3\left(\frac{\eta(3\tau)}{\eta(\tau)}\right)^6} \tag{5.1}$$

after the same deck convention. In the h-coordinate, the two noncuspidal cubic boundary values below are especially simple:

$$T = -27 \Longleftrightarrow h = -27, \qquad T = 729/5 \Longleftrightarrow h = 5.$$

It remains to compare compactifications. A direct singularity calculation for the six-variable cubic equation gives

cubic value	T-value	degeneration
$t = \infty$	$T = \infty$	six singular points
$t = 0$	$T = 0$	ten singular points
$t^2 = -1/3$	$T = -27$	five A_2-points
$t^2 = 9/5$	$T = 729/5$	chordal cubic

Only $T = 0$ and $T = \infty$ are cusps of $X_0(3)$. The marked smooth cubic base is therefore

$$\mathbf{P}^1_t \setminus \{0, \infty,\ t^2 = -1/3,\ t^2 = 9/5\},$$

and its unmarked image is

$$X_0(3) \setminus \{0, \infty, -27, 729/5\}.$$

On these opens, $t \mapsto T = 81t^2$ is finite *étale*. Its compactification ramifies only over the two modular cusps. This proves the boundary assertion of Theorem 1.1 and prevents a misleading identification of every cubic degeneration with a cusp.

6 Conclusion and scope

The object behind the five principal kernels is the S_3-torsor of ordered nonzero two-torsion on the actual relative norm axis. Its two proper transitive quotient sets are

$$S_3/C_2 \quad \text{(3 sheets)}, \qquad S_3/A_3 \quad \text{(2 sheets)}.$$

Thus the golden pair contains no independent monodromy datum: it is the discriminant orientation of the rational root triple. The familiar decomposition $\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}$ is the fibrewise form of this global resolvent theorem.

The modular identification here is deliberately limited. It neither turns the five principal kernels into ordinary elliptic isogenies nor promotes the whole compactified cubic pencil to a modular stack. The additional cubic boundary points at $T = -27$ and $729/5$ are precisely where that stronger claim would fail. Related level-three period maps occur for the Winger pencil [2, Theorem 1.1 and Theorem 5.2]; that is a distinct family and is not used to identify the cubic base. The contribution here is the combination of the explicit cubic period coordinate, the polarized actual-axis comparison, and the principal-kernel local system into one resolvent diagram.

Reproducibility. The directory `verification/` contains a deterministic SymPy certificate for (3.1)–(4.5) and (4.6), an independent enumeration of the relevant subgroups modulo six, their exact expected outputs, and a SHA-256 manifest. The paper's `make check` target replays both checks, verifies the manifest, lints the source, builds the PDF, and rejects TeX warnings.

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